Question1: Multiple Choice Question (15) Minutes [10] / 10 Marks

1	2	3	4	5	6	7	8	9	10
B	C	D	D	B	B	A	B	A	A

1. Which of the following big O-notations has constant order of growth?

   A. $O(n)$
   B. $O(1)$
   C. $O(n^2)$
   D. $O(\log n)$

2. The big O-notation of the expression $f(n) = 2^n + n^n + 6n^2 + 3n$ is:

   A. $O(2^n)$
   B. $O(n^2)$
   C. $O(n^n)$
   D. None of the above

3. Heapsort procedure sorts an array in place and has the following complexity:

   A. $O(n)$
   B. $O(\log n)$
   C. $O(n^2)$
   D. None of the above

   $n \log n$

4. What is the asymptotic relationship between $\log(n^{\log 6})$ and $\log(6^{\log n})$?

   A. $\log(n^{\log 6})$ is $O(\log(6^{\log n}))$
   B. $\log(n^{\log 6})$ is $\Theta(\log(6^{\log n}))$
   C. $\log(n^{\log 6})$ is $\Omega(\log(6^{\log n}))$
   D. All of the above.

   $\log n^{\log 6} \rightarrow \log 6 \log n$
   $6^{\log n} \Rightarrow n^{\log 6} = n^{\log 6}$

5. A sorting technique is called stable if it:

   A. Takes $O(n \log n)$ times
   B. Maintains the relative order of occurrence of non-distinct elements
   C. Uses divide-and-conquer paradigm
   D. Takes $O(n)$ space

Question 3: Recurrence (20) Minutes [4] / 4 Marks

$T(1)$ $i = \log_4 n$

1. Use the recursion tree method to solve the questions for the following recurrence:

$$T(n) = \begin{cases} \Theta(1) & \text{if } n = 1 \\ T(n/4) + n & \text{if } n > 1 \end{cases}$$

[2.5 marks]

i. How many levels in the recursion tree? **[0.5 mark]**

from 0 to $\log_4 n$ $\log_4 n + 1$ levels

ii. Determine the total cost of subproblems in ***level i*** (each level except the last one). **[0.5 mark]**

total cost at $i = \frac{n}{4^i}$

iii. Determine the total cost of subproblems in the last level (leaves). **[0.5 mark]**

total cost in the last level $= T(1) \times \frac{n}{4^{\log_4 n}} = T(1) \times \frac{n}{n} = T(1)$ constant

last level

iv. Calculate the total cost $T(n)$ of all levels, i.e., the total running time cost. Then represent T(n) using Big-Oh notation (O-notation). Show your work. **[1 mark]**

$$\text{Total cost} = \sum_{i=0}^{\log_4 n - 1} \frac{n}{4^i} * T(1) = n \sum_{i=0}^{\log_4 n - 1} \left(\frac{1}{4}\right)^i + T(1)$$

Since $n \sum_{i=0}^{\infty} \left(\frac{1}{4}\right)^i \geq n \sum_{i=0}^{\log_4 n - 1} \left(\frac{1}{4}\right)^i$ and $n \sum_{i=0}^{\infty} \left(\frac{1}{4}\right)^i = n\left(\frac{1}{1 - \frac{1}{4}}\right)$

Then $T(n) = n\left(\frac{1}{1 - \frac{1}{4}}\right) + T(1) = \frac{4}{3}n + T(1)$ → C

$T(n) = O(n)$

Question 2: **Fundamentals of Algorithms Analysis** **(15) Minutes [3.6]** **/ 5 Marks**

1- Consider the following running time function $T(n) = 4n^2 + 3n$
What is the Θ-notation of $T(n)$? What are the constants c_1 and c_2? **[2 marks]**

O-notation

$4n^2+3n \leq 4n^2+3n^2$

$\leq 7n^2$

$c_1=7$, $n_0=1$

$\therefore T(n)=O(n^2)$

Ω-notation

$4n^2+3n \geq 4n^2$

$\geq 4n^2$

$c_2=4$, $n_0=1$

$\therefore T(n)=\Omega(n^2) \Rightarrow$ Then $T(n)=\Theta(n^2)$

2- List the functions below from lowest to the highest order (growth wise). **[3 marks]**

$10^{\sqrt{n}}$	$n^{\log n}$	2^{100}	$\sqrt{n^n}$

Lowest Growth *Highest Growth*

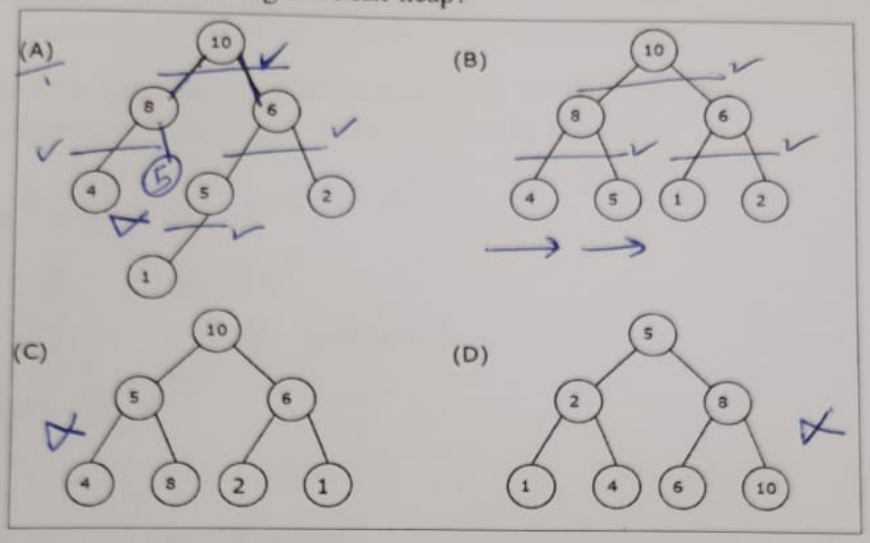
2^{100}	$10^{\sqrt{n}}$	$n^{\log n}$	$\sqrt{n^n}$

$\log n^{\log n}$ $\log n^{\frac{n}{2}}$

$\log n \cdot \log n \leq \frac{n}{2} \cdot \log n$

3.5

6. A Max-heap is a heap where the value of each parent is greater than or equal to the values of its children. Which of the following is a Max-heap?



A. (A)
B. (B)
C. (C)
D. (D)

7. The running time of Merge Sort in the worst case is:
A. $O(n \log n)$
B. $O(n^2)$
C. $O(1)$
D. $O(n)$

$a=1 \quad b=2 \quad f(n)=1$

8. Consider the recurrence: $T(n)=T(n/2)+1$. Using the Master method, choose the correct answer:
A. Apply case 1 and $T(n)=\Theta(1)$
B. Apply case 2 and $T(n)=\Theta(\log n)$
C. Apply case 3 and $T(n)=\Theta(1)$
D. Master method cannot be used.

$1 \quad n^{\log_b 1} \quad f(n)=\Theta(n^{\log_b a})$
$1 \quad n^0 \quad \therefore T(n)=\Theta(n^{\log_b a} \times \log_2 n)$
$1=1$

9. Suppose you have the following recurrence relation $T(n)=4T\left(\frac{n}{6}\right)+n$ then the number of levels in the recursion tree is:
A. $\log_6(n)+1$
B. $\log_4(n)+1$
C. $\log_2(n)+1$

$\log_6 n+1$

10. For any two functions $f(n)$ and $g(n)$, if $f(n)=\Theta(g(n))$ then $f(n)=O(g(n))$ and $f(n)=\Omega(g(n))$.
A. True
B. False

$f(n)=g(n) \quad f(n) \leq g(n) \quad f(n) \geq g(n)$

$n^2=n^2$

2. **Use the Master Method to give the tight upper bound (big-Oh) for the following recurrence:**

$$T(n) = 3\,T(n/5) + (\log n)^2$$ **[1.5 marks]**

$a = 3 \quad b = 5 \quad f(n) = (\log n)^2$

$(\log n)^2 \qquad n^{\log_5 3}$

$(\log n)^2 \leq n^{0.68}$

$f(n) = O(n^{\log_b a})$

$(\log n)^2 \leq 2^{n^{0.68}}$

$2^{(\log n)^2} \leq 2^{(2^{0.68n})}$

$\log n \leq (1.6)^n$

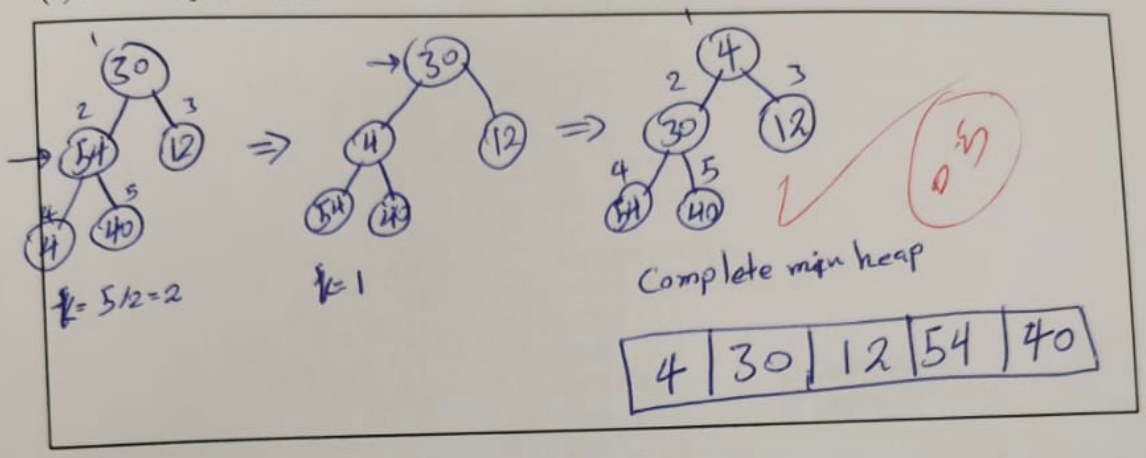
Case 1:

$\therefore T(n) = \Theta(n^{\log_b a}) = \Theta(n^{\log_5 3}) = O(n^{0.68})$

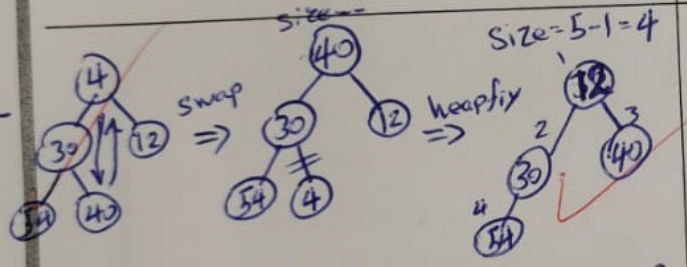
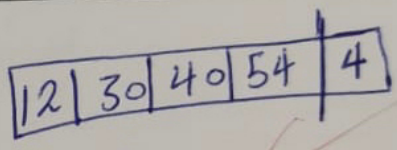
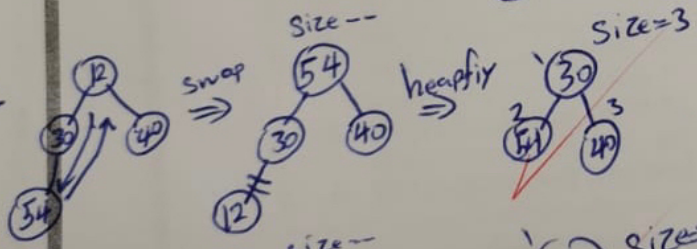
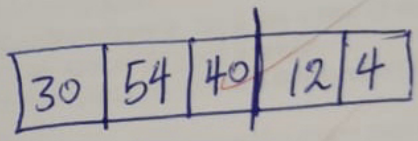
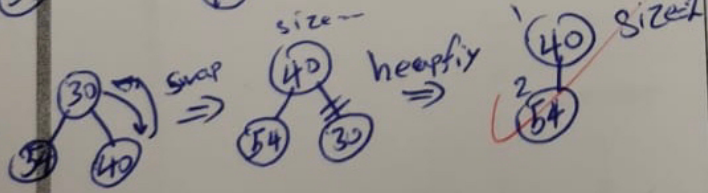
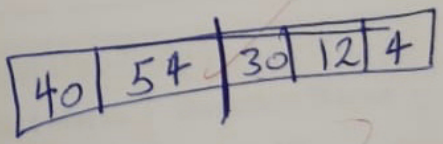
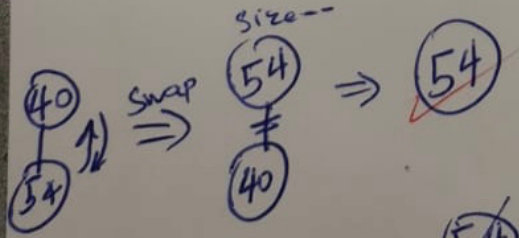
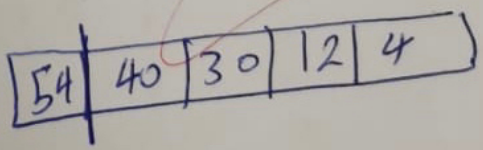
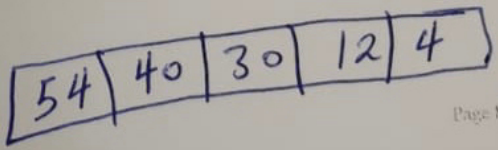
$O(n^{0.68})$

2. Using Min-Heap, apply the heapsort algorithm on the array A= {30, 54, 12, 4, 40}. Show the sorting details by (a) draw the min heap that represents array A and (b) draw each sorting step by showing the tree and the corresponding array.

(a) Min-Heap: **[0.5 mark]**



(b) The heapsort algorithm steps (4 steps): **[0.5 mark for each step]**

Tree	Array
1- 	
2- 	
3- 	
4- 	
	

Question 4: Sorting **(30) Minutes [6] / 6 Marks**

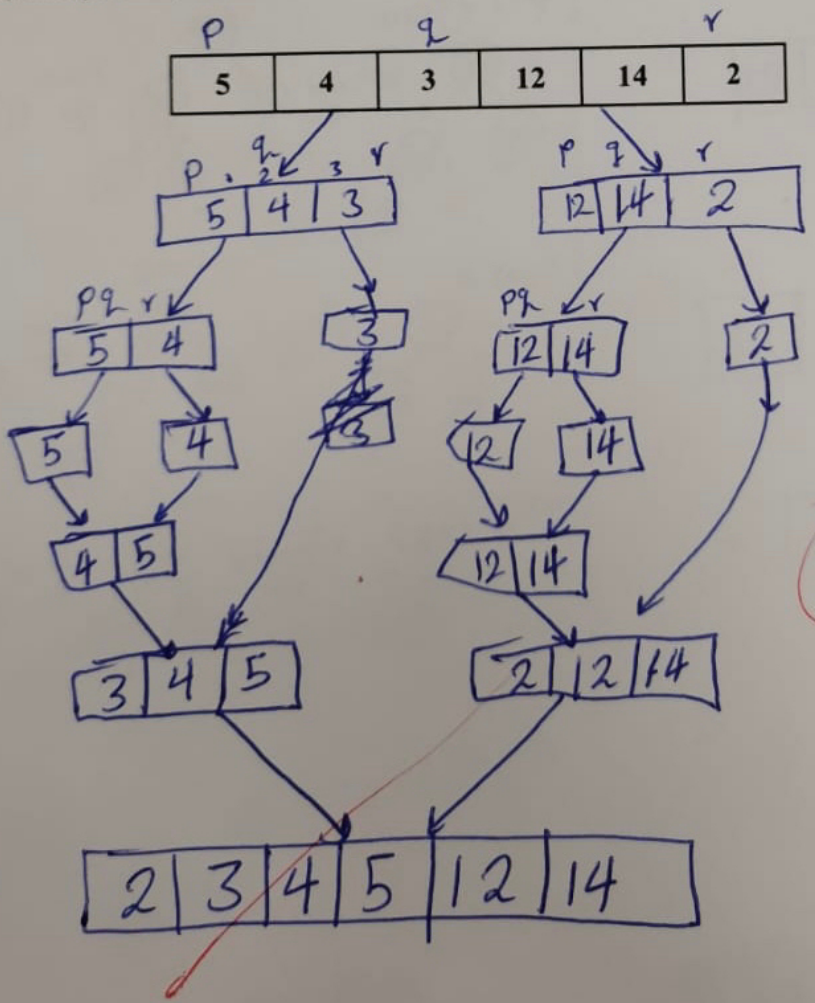
1. Illustrate (by drawing) the results and each iteration of sorting the array A={5, 4, 3, 12, 14, 2} using (a) Insertion sort and (b) Merge sort algorithms:

(a) Insertion sort **[1.5 marks]**

1st iteration	5	4	3	12	14	2
2nd iteration	4	5	3	12	14	2
3rd iteration	3	4	5	12	14	2
4th iteration	3	4	5	12	14	2
5th iteration	3	4	5	12	14	2
6th iteration	2	3	4	5	12	14

1.5

(b) Merge sort (show results after the combine step) **[2 marks]**



2